Swati Aggrawal

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PROFESSIONAL SUMMARY

MSc Computer Science graduate from Sapienza University of Rome specializing in Artificial Intelligence and Machine Learning. IEEE published researcher with hands-on experience developing machine learning, NLP, and Generative AI applications using Python, PyTorch, FastAPI, Docker, AWS, and Git. Experienced in building end-to-end ML pipelines, document intelligence systems, REST APIs, and AI-powered backend services. Passionate about applying AI to solve real-world engineering problems.

SKILLS

- **Programming :** Python, SQL, C++, Bash
- **Machine Learning & AI :** PyTorch, TensorFlow, Keras, Scikit-Learn, Deep Learning, Machine Learning, Computer Vision, Vision Transformers, Continual Learning, Transfer Learning, Hugging Face Transformers
- **NLP & Generative AI :** Generative AI, LLMs, Prompt Engineering, NLP, Text Processing, Embeddings, Vector Embeddings, Semantic Search, Retrieval-Augmented Generation (RAG), Sentence Transformers
- **Backend & APIs :** FastAPI, REST APIs
- **Cloud & DevOps :** Docker, AWS, ECS, CloudWatch, Git, Linux

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Computer Vision, Deep Learning and Applied AI, Natural Language Processing, Machine Learning, Distributed Systems
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Selected for a competitive international master's program.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

RESEARCH EXPERIENCE

Continual Learning for Computer Vision (Master's Thesis) **Nov 2025**

- Evaluated pre-trained Vision Transformer (ViT) models under continual learning settings using Python and PyTorch.
- Designed and implemented reproducible machine learning pipelines for data preprocessing, model training, evaluation, and performance benchmarking.
- Conducted comparative analysis of multiple continual learning strategies using standard evaluation metrics to measure model accuracy and catastrophic forgetting.
- Automated experiment execution and performance tracking to ensure reproducibility across sequential learning tasks.
- Optimized training workflows through systematic hyperparameter tuning and performance evaluation.
- Analyzed experimental results, documented findings, and presented insights to support future model improvements.
- **Technologies:** Python, PyTorch, Vision Transformers, Computer Vision, Git

Machine Learning for Air Pollution [IEEE Publication] **May 2023**

- Published IEEE conference paper evaluating machine learning approaches for environmental prediction and monitoring.
- Analyzed Random Forest, SVM, Genetic Algorithm, LS-SVM, Decision Trees, and Artificial Neural Networks across multiple air quality datasets.
- Performed comparative evaluation of model performance and identified opportunities for future research and real-time monitoring applications.
- Presented scientific findings through structured research documentation and publication.

PROJECTS

DocDelta AI-Powered Document Intelligence & Comparison Platform

- Developed an AI-powered document intelligence platform using Sentence Transformers, **vector embeddings**, and **semantic search** for document comparison and change detection.
- Built document chunking, semantic matching, and change detection pipelines for comparing multiple document versions.
- Built REST APIs using FastAPI for automated document analysis, **retrieval**, and semantic change detection workflows.
- Implemented embedding generation and retrieval workflows for semantic document search.
- Containerized the application using Docker and deployed it on AWS ECS Fargate with CloudWatch monitoring.
- Developed clean, modular, and reusable Python code following software engineering best practices.
- Implemented embedding generation and semantic retrieval workflows that form the foundation for Retrieval-Augmented Generation (RAG) applications.
- **Technologies:** Python, FastAPI, Sentence Transformers, Docker, AWS ECS Fargate, CloudWatch, REST APIs, Git

Metal Defect Detection using Computer Vision

- Developed and evaluated computer vision models for automated metal surface defect detection using the NEU Metal Surface Defects dataset.
- Designed a comparative study of CNN, Support Vector Machine (SVM), and Random Forest approaches to analyze the effectiveness of deep learning versus traditional machine learning techniques for industrial inspection.
- Implemented image preprocessing, data augmentation, feature extraction, and model training pipelines using TensorFlow, Keras, NumPy, and Scikit-learn.
- Performed systematic model evaluation using accuracy metrics, confusion matrices, and class-level performance analysis across multiple defect categories.
- Investigated the applicability of computer vision techniques for automated visual inspection, defect classification, and quality-control systems in manufacturing environments.
- Compared deep learning and traditional machine learning approaches to identify trade-offs between accuracy, computational cost, and deployment feasibility.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Scikit-Learn, Git

Lossy Text Compressor [Semantic NLP Pipeline]

- Investigated semantic text compression using transformer embeddings, quantization, and reconstruction techniques.
- Evaluated reconstruction quality using cosine similarity and ROUGE metrics.
- Developed an NLP pipeline for **text processing**, transformer-based **embedding generation**, semantic compression, and reconstruction.
- Evaluated embedding quality using cosine similarity and ROUGE metrics.
- Investigated semantic preservation across multiple quantization levels.
- Evaluated semantic preservation using vector embeddings, cosine similarity, and ROUGE metrics across multiple quantization levels..
- **Technologies:** Python, PyTorch, Transformer models, Git